

Primary and Metastatic Pancreatic Cancer Cells Exhibit Differential Migratory Potentials

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Objectives: Pancreatic ductal adenocarcinoma (PDAC) is characterized by early metastatic spread in more than 50% of patients. In this study, we sought to understand the migratory properties of (non)metastatic PDAC cells and determine whether the migration of cancer stem cell (CSC) populations accounts for the aggressive nature of this disease.

Methods: The migratory abilities of primary and metastatic PDAC cell lines were investigated using a microfluidic device and time-lapse photography. The velocity, time of delay of mobilization, and number of migratory cells were analyzed. Cancer stem cell subpopulations were isolated by fluorescence-activated cell sorting and their migratory properties compared with their non-CSC counterparts.

Results: Primary cancer cells exhibited higher velocities, greater number of migratory cells, and a shorter time of delay of mobilization in comparison to metastatic cell lines. Characterization of CSC populations revealed primary PDAC cell lines were composed of fewer CD133⁺ and CD24⁺CD44⁺ CSC subpopulations than metastatic cells. Moreover, migratory analysis of CSC subpopulations revealed lower velocities, fewer migratory cells, and a greater time of delay of mobilization than non-CSC.

Conclusions: Primary cancer cells demonstrate enhanced migratory abilities in comparison to metastatic PDAC cells. Those differences may result from lower CSC subpopulations in primary cells because CSC populations demonstrated impaired migratory abilities in contrast to non-CSC.

Key Words: CD133, metastasis, migration, pancreatic cancer, stem cells (*Pancreas* 2020;49: 128–134)

Pancreatic ductal adenocarcinoma (PDAC) is an aggressive malignancy associated with poor outcomes.¹ As a result, PDAC is on pace to become the second leading cause of cancer-related deaths by 2030.² Early metastasis and rapid local tumor growth are hallmarks of PDAC.³ Prior studies have indicated that only a small subset of highly plastic cancer cells with stem cell-like properties, cancer

stem cells (CSCs), are able to disseminate and form metastasis.^{4–7} Moreover, having the property of self-renewal, CSCs guarantee the long-term maintenance of tumor cells in a quiescent stage.^{8,9} However, the origin of CSCs is not completely understood, and whether they derive from mutated stem cells or non-CSC hybrid cells remains elusive.^{10,11} Recent studies have also demonstrated that the hierarchical lineage of CSCs is not unidirectional, and highly plastic non-CSCs are able to initiate tumor dissemination and replenish the CSC pool.^{12,13}

Multiple CSC markers, including CD133, ALDH1, CD24, CD44, and epithelial-specific antigen (ESA), have been employed to discriminate between different subpopulations in PDAC.^{14–16} Increased expression patterns of CD133⁺ and CD24⁺CD44⁺ESA⁺ markers have been associated with a poorer outcome in PDAC.^{17,18} Employing PDAC cell lines, it was shown that the different migratory abilities of CSC and non-CSC populations represent a key mechanism in PDAC progression.¹⁹ Moreover, CD133⁺ cells coexpressing CXCR4 drive the ability to metastasize, and cells expressing both markers are more frequently detected at the invasive front of tumors.¹⁵

In this study, we investigated different migratory abilities of a large number of cell lines derived from primary and metastatic tumors. Employing a microfluidic device that allows for the examination of cell migration properties at a single-cell resolution allowed for the identification of unique migration patterns in both primary- and metastatic-derived cell lines and CSC and non-CSC populations.

MATERIALS AND METHODS

Tumor Cell Lines and Cell Culture

Pancreatic ductal adenocarcinoma cell lines BxPc-3, Capan-2, MIA PaCa-2, MPANC-96, PANC-1, PaTu 8902, AsPC-1, Capan-1, COLO 357, HPAF-II, KP-1NL, PaTu 8988, and SW 1990 were obtained from the American Type Culture Collection (ATCC, Manassas, Va) and maintained in the recommended culture media. Normal pancreatic duct cell line HPDE6E7 (HPDE) was kindly provided by Dr Ming-Sound Tsao (Ontario Cancer Institute, Toronto, Canada) and maintained in keratinocyte serum-free medium supplemented by epidermal growth factor and bovine pituitary extract (Gibco-BRL, Grand Island, NY). Patient-derived xenograft (PDX) tumors were generated after the approval and in accordance with the requirements of the institutional review board of Massachusetts General Hospital, Boston, as previously described.²⁰ Cell lines derived from these PDX models (MGH-609, MGH-617, MGH-1108, MGH-1275, MGH-1312) were grown in Dulbecco modified eagle medium:F-12 (50:50) supplemented with 10% fetal bovine serum (FBS).

Fluorescence-Activated Cell Sorting

Cell suspensions were counted and transferred to a 5-mL tube, washed twice with Hanks balanced salt solution (HBSS)

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containing 2% heat-inactivated FBS, and resuspended in HBSS with 2% FBS at a concentration of 10^6 per 100 μL . Sandoglobulin solution (1 mg/mL) was then added to the sample at a dilution of 1:20 before incubation for 20 minutes on ice. The sample was then washed twice with HBSS/2% FBS and resuspended in HBSS/2% FBS. Antibodies were added and incubated for 20 minutes on ice, and the samples were washed twice with HBSS/2% FBS. To characterize pancreatic CSCs, the following antibodies were used: anti-CD44 allophycocyanin, anti-CD24 PE, CD133/1-allophycocyanin, or CD133-PE (Miltenyi, Bergisch-Gladbach, Germany) and anti-ESA-fluorescein isothiocyanate (Biomedica, Foster City, Calif), each at a dilution of 1:40. Flow cytometry was done using a FACS Aria (BD Immunocytometry Systems, Franklin Lakes, NJ). Side-scatter and forward-scatter profiles were used to eliminate cell doublets. Cells were routinely sorted twice, and the cells were reanalyzed for purity, which was routinely greater than 97%.

Migration Assays Using a Microfluidic Device

The migration of PDAC cell lines was analyzed in a microfluidic device containing microcapillary arrays fabricated as described previously.^{21,22} Briefly, devices were coated with collagen IV immediately after bonding and while the PDMS (polydimethylsiloxane; Dow Corning, Midland, Mich) was still hydrophilic. Within 2 hours after device fabrication, 3 μL of cell suspension was loaded in the device at 10^6 to 10 cells/mL, and individual wells on the plate were filled with 3 mL of media (RPMI/10% FCS), completely covering the microfluidic devices. The multiwell plate was mounted on the automated stage of a Zeiss Axiovert microscope (Carl Zeiss Microscopy, Jena, Germany) equipped with an environmental chamber set at 37°C and 5% CO₂. Cells were imaged using a 10 $\times$ objective and phase contrast with individual frames acquired from 3 different locations of each device every 15 minutes up to 96 hours. Moving cells were tracked manually using Image J software (version 1.46; National Institutes of Health, Bethesda, Md). Motility metrics included velocity of the cells ($\mu\text{m}/\text{h}$), time of delay of mobilization (hours), and number of cells mobilized.

Statistical Analysis

Statistical analyses were performed using GraphPad Prism 7 Software (GraphPad Software Inc, San Diego, Calif). Continuous

variables were analyzed using the Student *t* test and analysis of variance. Data are presented as mean $\pm$ standard error of the mean (SEM), and the statistical significance was accepted at 2-sided *P* < 0.05.

RESULTS

Primary PDAC Cells Demonstrate Increased Migratory Abilities

Because metastasis and early tumor dissemination are key features in pancreatic cancer, we wanted to investigate the migratory abilities of PDAC and normal ductal epithelial cells (HPDE). Therefore, PANC-1 and HPDE cells were seeded in a microfluidic device that allows for the analyses of their migratory properties at a single-cell resolution.²¹ In contrast to HPDE cells, PANC-1 cells were able to efficiently undergo spontaneous migration (Fig. 1A and Supplemental Videos 1 and 2, <http://links.lww.com/MPA/A761>, <http://links.lww.com/MPA/A762>). Quantitative analysis of single-cell distances in the device revealed a much higher migratory distance in the microfluidic capillaries of PANC-1 cells in comparison to HPDE cells (Fig. 1B), suggesting increased migratory ability of malignant cells.

To examine the heterogeneity of the migratory properties of PDAC cells, cell lines derived from primary tumors (*n* = 6) and metastatic lesions (*n* = 7) were placed as single-cell suspensions in the microfluidic device. In line with the findings above, spontaneous movement was observed in all PDAC cell lines. Moreover, a significantly higher number of cells migrated from cell lines derived from primary PDAC tumors in comparison to those derived from metastatic tumors (Fig. 2A). On average, 2836 cells (56.7%) were able to migrate within 96 hours in the primary PDAC cell line group, whereas only 1414 cells (28.3%) migrated in metastatic PDAC cell line group (*P* = 0.002). In primary PDAC cells, the highest number of cells that underwent spontaneous movement was detected in BxPC-3 and MIA PaCa-2 cells with 3750 and 3200 cells, respectively. In comparison, Capan-2 and PaTu 8902 showed the lowest number of migrated cells with fewer than 2500 cells. In the metastatic group, KP-NL1 had the highest number of migrated cells with 2000 cells, whereas COLO 357 cells exhibited a low migratory impulse with fewer than 1000 cells.

To gain insight into the mechanisms responsible for these migration differences, we investigated the time required for cells to

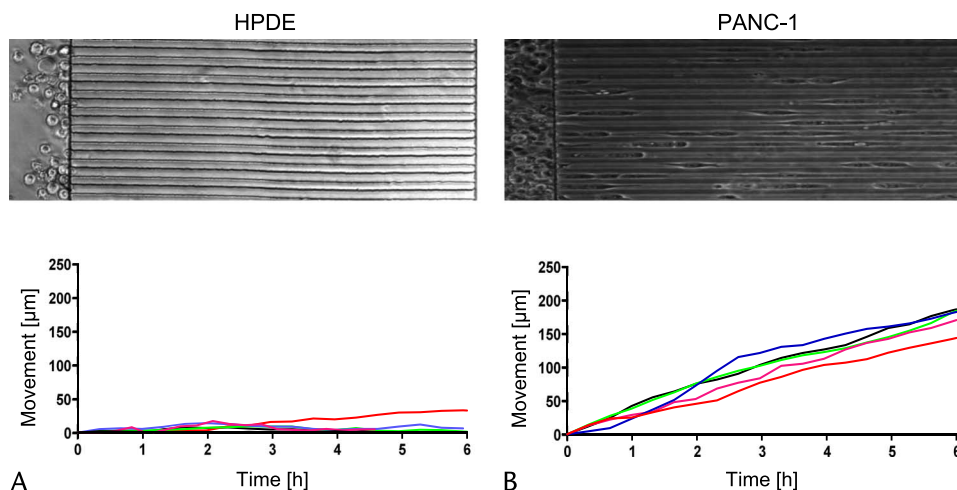


FIGURE 1. Pancreatic ductal adenocarcinoma cells have increased migratory abilities than nonmalignant ductal epithelial cells. Representative images of human pancreatic ductal cells (HPDE) and PANC-1 cells in the microfluidic device on a single-cell level (A and B; upper panels). The distance of migration (μm) of HPDE cells (A, lower panel) and PANC-1 cells (B, lower panel) within the channels of the microfluidic device was measured over 6 hours. The average distance of 200 cells is plotted for each cell line. **Editor's note:** A color image accompanies the online version of this article.

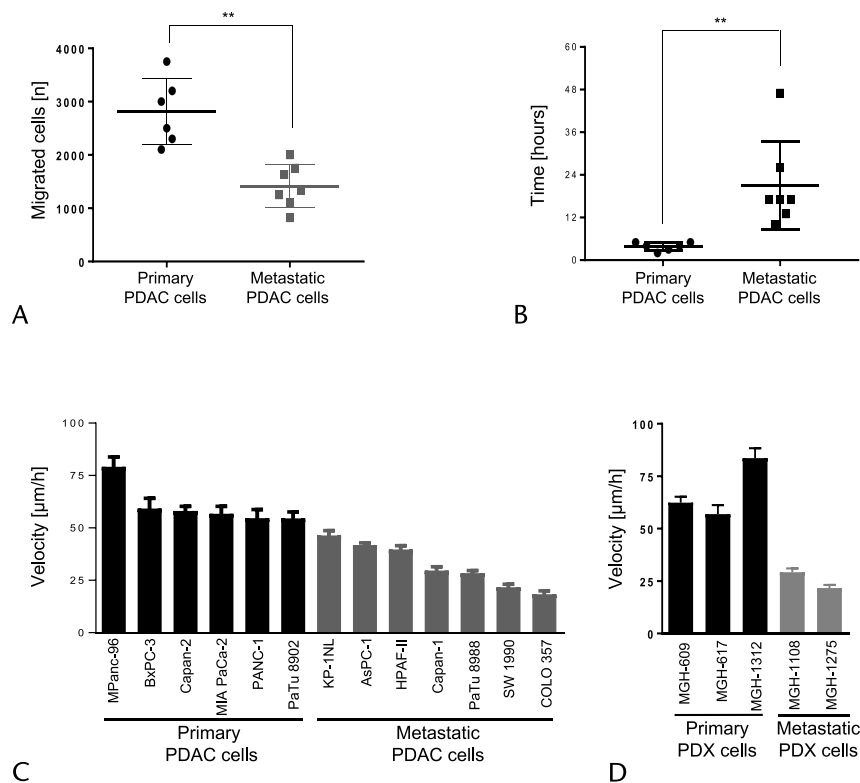


FIGURE 2. Primary and metastatic PDAC cells exhibit different migration properties. The numbers of cells undergoing spontaneous migration in primary and metastatic PDAC cell lines were quantitated over 96 hours (A). The time of delay to mobilization (hours) for primary and metastatic cell lines to undergo spontaneous migration in the microfluidic device was measured (B). Velocities, expressed as the time-dependent migration distance ($\mu\text{m}/\text{h}$), were assessed in 6 primary (black bars) and 7 metastatic (gray bars) PDAC cells lines (C). The velocities ($\mu\text{m}/\text{h}$) of PDX-derived primary (black bars) and metastatic (gray bars) cells were measured (D). Each point (B) represents the average migration property of 2000 PDAC cells. **Changes in migration properties $P < 0.01$.

begin to migrate after seeding in the microfluidic device (Fig. 2B). Pancreatic ductal adenocarcinoma cell lines exhibited a broad range in delay to mobilization, ranging from 2.5 hours (PANC-1) to 47 hours (SW 1990). Overall, primary PDAC cells had a significantly shorter time of delay to mobilization compared with metastatic PDAC cells. Primary PDAC cells began to migrate in the microfluidic device after a mean of 4.2 hours ($\text{SEM} \pm 0.61$ hour), whereas each of the metastatic PDAC cell lines failed to demonstrate any migratory impulse during this period. Metastatic cell lines exhibited a 5-fold greater delay of mobilization with a mean delay of 21 hours ($\text{SEM} \pm 4.72$ hours; $P = 0.004$).

To further explore the migratory speed on a single-cell level, we recorded the velocities for the individual cell lines over 96 hours (Table 1). Measuring the velocities of at least 2000 cells for each cell line revealed that primary PDAC cells exhibited 2-fold higher velocities than metastatic PDAC cells, with a mean velocity of $50.29 \mu\text{m}/\text{h}$ for primary PDAC cells compared with $24.71 \mu\text{m}/\text{h}$ for metastatic PDAC cells (Fig. 2C; $P < 0.01$).

Many of the PDAC cell lines employed in this study were derived more than 3 decades ago, and therefore some of the intrinsic biological characteristics of the cells may have been altered by extensive passaging. To examine whether the migratory differences between primary and metastatic PDAC cells exist in low-passage cell lines, we performed similar studies with 3 primary and 2 metastatic cell lines derived from PDX tumors. Consistent with the results obtained with commercial cell lines, PDX-derived cell lines from primary tumors showed significantly higher velocities than metastatic PDX-derived cell lines (Fig. 2D; $P < 0.01$). The mean

velocities were $67.62 \mu\text{m}/\text{h}$ in primary cells and $25.43 \mu\text{m}/\text{h}$ in metastatic cells. Moreover, the highest velocity of all cell lines we investigated was in MGH-1312, with a migratory speed of $83 \mu\text{m}/\text{h}$. Taken together, primary PDAC cells had a significantly increased migratory capacity and velocity compared with metastatic PDAC

TABLE 1. Migratory Abilities of PDAC Cell Lines

Cell Line	Origin	Velocity, $\mu\text{m}/\text{h}$	Time of Delay, h	No. Cells, n
MPanc-96	P	79.20	2	3200
BxPC-3	P	59.25	3	3750
Capan-2	P	58.08	5	2300
MIA PaCa-2	P	56.71	4	3000
PANC-1	P	54.61	4	2500
PaTu 8902	P	54.51	5	2100
KP-1NL	M	46.38	13	2000
AsPC-1	M	41.89	10	1633
HPAF-II	M	39.73	26	1750
Capan-1	M	29.66	17	1250
PaTu 8988	M	28.43	17	1340
SW 1990	M	21.66	26	1100
COLO 357	M	18.30	47	820

P indicates primary PDAC cell line; M, metastatic PDAC cell line.

cells, which suggests a more aggressive biology of primary than metastatic cancer cells.

Different CSC Populations Between Primary and Metastatic PDAC Cells

Previous reports have suggested that primary and metastatic cell lines express distinct CSC populations.^{23,24} Therefore, we investigated whether the different migratory abilities between primary and metastatic cell lines are related to different CSC subpopulations. Fluorescence-activated cell sorting (FACS) analyses were performed for ESA, CD24, CD44, and CD133 cell surface markers to identify different CSC subpopulations in primary and metastatic

PDAC cell lines. Almost 100% of the cells were positive for ESA, which was in contrast to markers for the CSC populations (Fig. 3 and Supplemental Fig. 1, <http://links.lww.com/MPA/A757>). More than 65% of cells in cultures of Capan-1, a metastatic PDAC cell line, were CD133⁺ESA⁺. Furthermore, CD24⁺ESA⁺ and CD44⁺ESA⁺ markers were found in 87% and 96% of Capan-1 cells, respectively (Fig. 3A). Of these cells, 81% were triple-positive for CD24⁺CD44⁺ESA⁺. In contrast to the high levels of CD133⁺ESA⁺ cells in Capan-1 cells, cultures of BxPC3, a primary PDAC cell line, contained less than 5% of CD133⁺ESA⁺ cells (Fig. 3B). In contrast, CD24⁺ESA⁺ and CD44⁺ESA⁺ markers were found in 93% and 98% of BxPc3 cells, respectively. Of these cells 83% were positive for the triple marker, CD24⁺CD44⁺ESA⁺. Next,

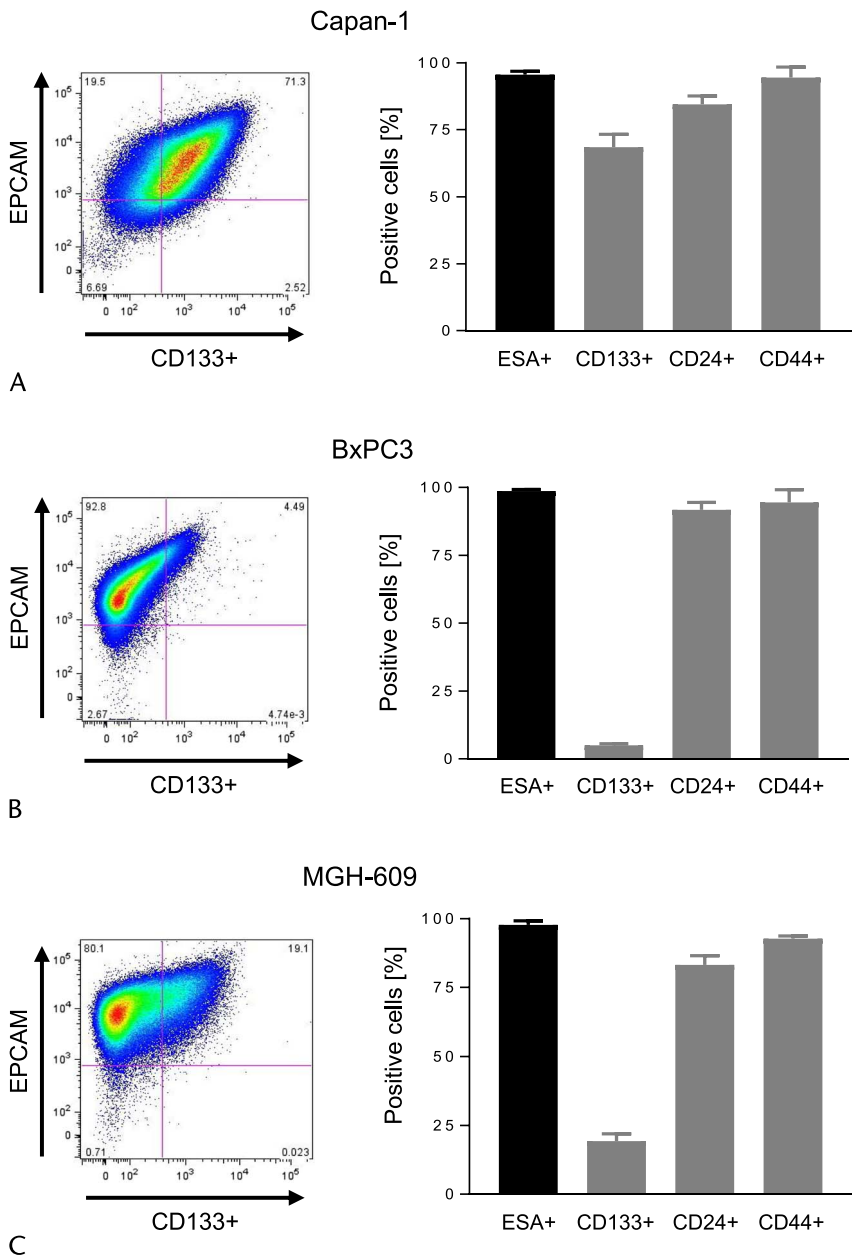


FIGURE 3. Heterogeneous CSC populations among PDAC cell lines. Representative FACS analyses of ESA⁺ and CD133⁺ subpopulations in the metastatic PDAC cell line Capan-1 (A, left), primary cell line BxPC3 (B, left), and the primary PDX-derived cell line MGH-609 (C, left). Percentages (standard deviation) of ESA⁺, CD133⁺, CD24⁺, and CD44⁺ cell populations from 3 independent FACS analysis are shown in the graphs on the right. **Editor's note:** A color image accompanies the online version of this article.

we analyzed our primary PDX-derived cell line MGH-609 to confirm these findings in our PDX-derived cell line model. The level for CD133⁺ESA⁺, CD24⁺ESA⁺, and CD44⁺ESA⁺ were 19%, 83%, and 92%, respectively (Fig. 3C). Thirty-one of the cells showed triple-positivity for CD24⁺CD44⁺ESA⁺. Further highlighting the differences in CSC populations between primary and metastatic cell lines, the percentage of cells expressing all 4 CSC markers (CD133⁺CD44⁺CD24⁺ESA⁺) in BxPC-3 and MGH-609 cells (1.9% and 5.9%) was much lower in comparison to Capan-1 cells (38%).

CSCs Show Lower Migratory Potentials Than Non-CSC

To directly compare the migratory abilities of CSC and non-CSC populations, we further investigated subpopulations derived from MGH-609 cells. Cancer stem cell and non-CSC populations were isolated by FACS and evaluated in the microfluidic device (Supplemental Videos 3 and 4, <http://links.lww.com/MPA/A759>, <http://links.lww.com/MPA/A758>). Cancer stem cells had a significantly lower percentage of cells that were capable of spontaneous migration compared with non-CSC. These effects were demonstrated for both CD133⁺ESA⁺ and CD24⁺CD44⁺ESA⁺ stem cell populations. Only 1.3% of the CD133⁺ESA⁺ cells were capable of a spontaneous migration, whereas 54.7% of non-CSCs migrated in the microfluidic device (Fig. 4A; $P < 0.001$). Similarly, 1.3% of CD24⁺CD44⁺ESA⁺ cells underwent a spontaneous migration versus 47.8% of the corresponding non-CSCs (Fig. 4D; $P < 0.001$).

Consistent with these results, the time of delay of mobilization was significantly longer for CSCs than non-CSCs (Fig. 4B). In the CD133⁺ESA⁺ population, we observed a delay of 45 hours (SEM ± 1.2 hours) compared with 3.3 hours in CD133⁺ESA⁺ cells (SEM ± 0.3 hour; $P < 0.001$). Although not reaching statistical significance, cells positive for the triple CSC marker

CD24⁺CD44⁺ESA⁺ exhibited a longer median delay to mobilization of 28 hours (SEM ± 5.4 hours) in comparison to 18 hours (SEM ± 4.06 hours) for the corresponding non-CSCs (Fig. 4D).

Finally, examining the individual velocities of these cells revealed that CSC subpopulations had significantly slower velocities than non-CSC populations (Figs. 4C, F). CD133⁺ESA⁺ exhibited a mean velocity of 6 $\mu\text{m/h}$ (SEM ± 0.5 hour) compared with 24 $\mu\text{m/h}$ (SEM ± 0.95 hour) for the corresponding non-CSCs ($P < 0.001$). Similarly, CD24⁺CD44⁺ESA⁺ exhibited a mean velocity of 5.9 $\mu\text{m/h}$ (SEM ± 1.1 hours), whereas the corresponding non-CSCs had a mean velocity of 35 $\mu\text{m/h}$ (SEM ± 1.8 hours; $P < 0.001$). Taken together, these results suggest that CSCs have reduced migratory properties compared with non-CSCs, which is consistent with the divergent migratory phenotypes observed in primary and metastatic cell lines.

DISCUSSION

Pancreatic ductal adenocarcinoma is a highly malignant neoplasia with a rapid local growth and early tumor dissemination. In the present study, we investigated the different migratory potentials of a large number of primary and metastatic cell lines, and we explored the migratory behavior of CSCs in comparison to non-CSC subpopulations. During the past decade, there have been many studies characterizing CSC markers in cancer.¹² The expressions of CD133⁺ and CD24⁺CD44⁺ESA⁺ surface proteins are the most common markers for CSCs, and their expression has been associated with poor prognosis and metastatic behavior in different cancer types.¹⁸ In PDAC, it was demonstrated that CD24⁺CD44⁺ESA⁺ cells form tumor spheres, whereas CD24⁺CD44⁺ESA⁺ cells did not.²⁵ Similar findings were reported for CD133⁺ cells that were capable to form tumors even with lower cell numbers.¹⁵ Most of these data were generated from in vitro or transplantation studies using PDAC cell lines that have undergone a selection

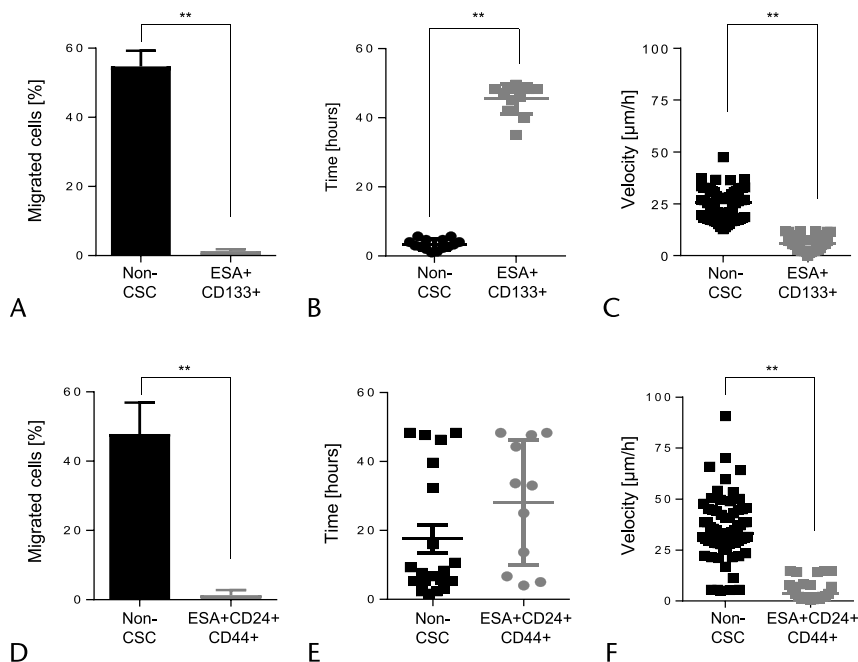


FIGURE 4. Lower migratory abilities in CSCs derived from primary PDX cells. Percentages of ESA⁺CD133⁺ (gray) and ESA⁺CD133⁻ (black) subpopulations that migrated over 96 hours in the microfluidic device (A). Time of delay to mobilization (hours) was measured for ESA⁺CD133⁺ and ESA⁺CD133⁻ cells after implantation in the device (B). Velocities ($\mu\text{m/h}$) of ESA⁺CD133⁺ and ESA⁺CD133⁻ were measured over 96 hours (C). ESA⁺CD24⁺CD44⁺ cells and the corresponding non-CSC populations were isolated by FACS, and the percentage of cells that migrated (D), time of delay to mobilization (E), and velocities (F) were defined. Each dot represents the average of 2000 PDAC cells (B, C, E, and F). **Changes in migration properties $P < 0.01$.

process under optimized cell culture conditions.^{20,26} As a result, the expression of CSC markers is differentially distributed among the substantial number of cell lines reassembling the intratumoral heterogeneity of PDAC.^{27,28} Therefore, it is important to understand the specific abilities of the established cell lines, including the migratory potential.

Using a microfluidic device, we demonstrated that primary PDAC cells showed a much higher migratory capacity than metastatic cells. These differences were observed for different migratory properties and revealed not only an increased velocity but also a shorter time to mobilization together with a higher number of cells that migrated in primary cell lines. Those PDAC cells are also more migratory and more accurate at navigating through mazes compared with other cell lines.²⁹ This would support the hypothesis that primary cells are more likely to dissociate from the primary tumor site and enter the blood or lymphovascular system to seed metastasis.

Further analyses of isolated CSC subpopulations demonstrated that CD133⁺ and CD24⁺CD44⁺ESA⁺ subpopulations had fewer migratory cells and a significantly lower migratory speed than non-CSC fractions. The divergent migratory phenotypes of the CSC and non-CSC populations reflected, in part, the differences in migratory behavior of primary and metastatic cell lines. Primary cells had smaller CSC subpopulations and a more aggressive migratory phenotype than cell lines of a metastatic origin.

While these findings do not exclude the ability of CSCs to disseminate and form metastasis, it seems that they are not the exclusive driver of migration in the cell lines. Other reports suggest an increased migratory ability by EMT for CD133⁺ subpopulations and demonstrated a more aggressive growth pattern in vitro.³⁰ Given the clusters of circulating tumor cells have been observed, one intriguing possibility is that CSCs use cell-cell contacts with non-CSCs to migrate to distant sites. Additionally, there is growing evidence that CSCs do not necessarily represent an entity, but rather a reversible CSC state.³¹ In gastrointestinal cancers, CSC-like cells arise de novo from non-CSC populations highlighting the plasticity of cancer cells.^{32,33} In glioblastoma, it has been shown that CD133⁺ cells were able to generate clones with tumorigenic capacity, regardless of other subpopulations.³⁴ Additionally, recent studies demonstrated that non-CSCs may have a transient proliferative capacity, which allows for the temporary implementation of metastasis.^{11,35} Our results would support a model in which primary tumor cells do not necessarily carry CSC markers when they enter the circulation but undergo a CSC-like phenotype and initiate distant tumor growth once they reach secondary organs.

Taken together, characterizing the migration of primary and metastatic cell lines at a single-cell resolution demonstrated primary cancer cells have a much more aggressive migratory phenotype than metastatic PDAC cells and isolated CSC subpopulations. These differences are related, at least in part, to a lower level of CSC subpopulations in primary cells and identify non-CSC populations with a strong migratory behavior than CSC. Together with recent findings that CSCs are not exclusively the sole driver for tumor dissemination, our study underlines the role of non-CSCs in the bidirectional plasticity of PDAC cells. The different migratory properties of these subpopulations will aid in understanding their contributions to the aggressive metastatic behavior of PDAC and may ultimately generate more effective therapies addressing those cancer cells.

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